

Upload the source code Program_11973.ino to the Arduino board by clicking on the Upload button. Before compiling and uploading the source code, do not forget to include ESP8266WiFi.h, DHT11.h, Adafruit,

and MQTT libraries from Library Manager.

NodeMCU Wi-Fi is connected to Adafruit.io open source cloud service to provide real-time data online. Message queue telemetry transport

(MQTT) is a protocol for device communication that Adafruit.io supports.

Steps to organise the dashboard on Adafruit.io IoT platform:

1. Open <https://io.adafruit.com> website. Create an account with a unique username. Note down this username as it will be used in Arduino code (Program_11973.ino) later.

2. Generate a key and note down this key. This is a long, unique identifier that you use to authenticate any device using your account. You get one key per account, but you can revoke and regenerate your key any time.

3. After creating an account, you need to manage your feeds for publishing on the dashboard as shown in Fig. 5. (Feeds is basically a set of data that you can read or write, as specified in the program, as per your application.)

In this project, we have used four moisture sensors (MS1 through MS4), and a temperature sensor (tmp). So, as an example, if this program used the following in the code, then it would require to create each feed as MS1 through MS4, tmp, LHT, etc to manage on dashboard:

```
Adafruit_MQTT_Publish photocel3 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/MS1");
Adafruit_MQTT_Publish photocel4 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/MS2");
Adafruit_MQTT_Publish photocel5 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/MS3");
Adafruit_MQTT_Publish photocel6 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/MS4");
Adafruit_MQTT_Publish photocel1 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/tmp");
Adafruit_MQTT_Publish photocel2 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/hum");
Adafruit_MQTT_Publish photocel7 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/LHT");
```

You need to add all feeds in Adafruit IO to display live data from sensors on dashboard as shown in Table 3.

More details regarding configuration with Adafruit IoT and Arduino are given in 'Configurations with Adafruit & Arduino' document file in the source code folder.

Now open PROGRAM_11973.ino source code, include all your own

Sensor	Pin Name	NodeMCU	74HC4067
Vibration sensor	A0(O/P)	D8(I/P)	-
Soil sensor 1-4	A0(O/P)	-	C0-C3(I/P)
LDR module	O/P	-	C4(I/P)
DHT11	DATA(O/P)	D6(I/P)	-
Relay drive	IN1(I/P) IN2(I/P)	D5(I/P) D7(I/P)	-
16x1 MUX	S0-S3(I/P) SIG(O/P)	D0-D3(O/P) A0(I/P)	-

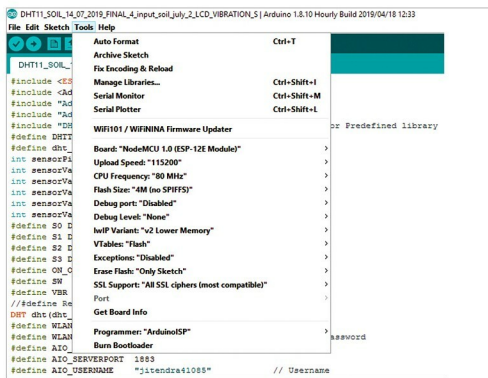


Fig. 4: Selection of board as NodeMCU 1.0 (ESP-12E module)

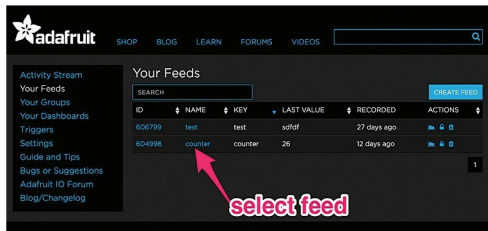


Fig. 5: Feeds window on Adafruit IoT platform